Fiscal Year:	FY 2018	**Task Last Updated:**	FY 12/28/2017
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Project Title:	Changes in the Neuroproteome Associated with HZE-Induced Impairment of Cognition		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	23507-1607	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 12/31/2018 per S. Monk/SR/LaRC (Ed., 1/11/18)		
Key Personnel Changes/Previous PI:	Dec 2016: Drs. Semmes and Dutta were removed from the project; proteomic analysis is now being conducted at UTMB (University of Texas Medical Branch) as contract work.		
COI Name (Institution):			
Grant/Contract No.:	NNX14AE73G		
Performance Goal No.:			
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Task Description:	One of NASA's biggest concerns about the long-term health of astronauts who go on deep space missions is the impact that High Z, High Energy (HZE) particles have on brain function (neurocognition). Data from our laboratory and others suggests that there is significant impairment of certain neurocognitive tasks (spatial memory and Executive function-attentional set shifting) following exposure to low HZE doses. The goal of this application is to determine the Threshold dose for the induction of HZE-induced spatial memory impairments (HISMI) or Attentional Set Shifting Impairments (HIASSI) following exposure to 56Fe, 48Ti, and 28Si particles. The proposed studies will also identify the changes in the proteome of the brain (neuroproteome) of rats that differ in their susceptibility to HISMI and HIASSI, which will provide further insight into the factors that lead to HISMI/HIASSI and perhaps more importantly, that prevent its emergence. Our underlying hypothesis is that HISMI and HIASSI arise as the direct result of HZE-induced changes in the neuroproteome. We also hypothesize that exposure to HZE species that have different track structures will result in different mechanisms of HZE-induced cognitive impairment (HICI). Collectively, these studies will give some insight into the underlying cause for HISMI and HIASSI. Our studies will thus address CNS (Central Nervous System) Gaps 1, 2, and 6, and we shall specifically focus on the following aims: Aim 1. Determine the Threshold dose for the induction of HISMI and HIASSI following exposure to 56Fe, 48Ti, and 28Si particles when delivered as a single dose. Aim 2. Identify changes in the neuroproteome that are associated with susceptibility or resistance to developing HISMI and HIASSI following exposure to 56Fe particles. Aim 3. Determine the mechanism of HISMI and HIASSI induced by HZE particles of differing LET (linear energy transfer). In Aim 1, socially mature (~6 month old) male Wistar rats will be irradiated with 56Fe, 48Ti, and 28Si particles (with incident energies of 600 MeV/nucleon). Rats will receive whole body HZE irradiation (< 15 cGy), and HISMI and HIASSI will be assessed at 3 months post irradiation. In Aim 2 and 3, the composition of the neuroproteome (hippocampus and selected regions of the prefrontal cortex) of irradiated rats that have “normal” cognitive performance or have developed HISMI or HIASSI will be established using an unbiased proteomic profiling approach. We shall use a label free differential protein profiling workflow on the Q-Exactive Orbitrap mass spectrometer. These studies will give considerable insight into the underlying cause for HZE-induced neurocognitive failure. The proposed studies will continue to define the minimum dose of HZE particles that will induce HISMI and HIASSI. Moreover, our studies will provide considerable insight into the underlying mechanism of HICI, and will identify prognostic biomarkers that could be translated to human studies to monitor the emergence of HICI. These studies may also help to develop appropriate countermeasures and help identify sensitive individuals, so that NASA’s medical staff can implement appropriate countermeasures to protect these at risk individuals.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These studies will give considerable insight into the underlying cause for HZE-induced neurocognitive failure. The proposed studies will continue to define the minimum dose of HZE particles that will induce HISMI and HIASSI. Moreover, our studies will provide considerable insight into the underlying mechanism of HICI, and will identify prognostic biomarkers that could be translated to human studies to monitor the emergence of HICI. These studies may also help to develop appropriate countermeasures and help identify sensitive individuals, so that NASA’s medical staff can implement appropriate countermeasures to protect these at risk individuals.
	Project Objectives 1. Identify the lowest Galactic Cosmic Radiation (GCR) dose that results in Attentional Set Shifting (ATSET) impairment. 2. Determine if there are LET-specific mechanisms of ATSET Impairment. 3. Identify changes in the neuroproteome that reflect the cognitive performance status of GCR-exposed animals. Project Approach: To better simulate the “clinical reality,” adult (proven breeder) male rats that have been pre-selected for good attentional set shifting (ATSET) performance, and who have been maintained on an exercise regimen are used in this study. Rats are exposed (primarily a single exposure) to <15 cGy of GCR ions (incident energy <600 MeV/n) and ATSET performance re-established at 3 months post exposure. The post-exposure ATSET test is reconfigured from the pre-exposure ATSET test, so that the rats have to “relearn” the associative clues, and importantly the post-exposure test uses associative clues in the first 6 paradigms that are “media”-based and not olfactory-based. These changes eliminate memory retention issues and also GCR-induced changes in olfaction. Regions of the cortex and basal forebrain (that regulated certain paradigms with the ATSET test) are recovered and subjected to proteomic analysis to identify some of the processes that may be responsible for the GCR-induced impairment of ATSET. Work conducted to date: To date ~500 rats that have been selected for good ATSET performance have been used to assess the ATSET performance after exposure to <10 cGy of 400 MeV/n 4He, 400 MeV/n 16O, 600 MeV/n 28Si, 600 MeV/n 48Ti, and 600 MeV/n 56Fe ions. In addition to accruing data that will help establish the lowest dose of GCR radiation that will impair ATSET, during this reporting period we have also determined whether GCR exposure impacts “established” working memory. Furthermore, our studies have focused on presenting our data in a format that can be readily used to calculate a Probabilistic Risk Analysis (PRA) for cognitive impairment, i.e., establishing the severity and frequency of severe ATSET impairment. While the ATSET test provides considerable insight into the problem-solving capacity of rats, it is limited in that it only

Task Progress:	measures the ability of rats to shift attention between a constrained set of options. Humans also have the ability to conduct creative problem using unconstrained cognitive flexibility. There has been a recent evolution of the ATSET test, the Unconstrained Cognitive flexibility (UCFlex) assay that requires creative problem solving [Hecht et al., 2014]. During this reporting period we have developed our own version of the UCFlex assay and have expanded our studies to include unconstrained cognitive flexibility performance. Research Highlights • GCR doses as low as 1 cGy lead to the impairment of ATSET performance, particularly in the tasks regulated by the mPFC and perirhinal regions. (To date we have not seen any measureable ATSET impairment by 16O ions- only 1 batch analyzed, another batch about to be rescreened in Feb 2018.) • However, imprinted memories are preserved following GCR exposure, suggesting that highly "entrained" skill sets may not be affected by GCR exposure. • The addition of the UCFlex test not only increases the spectrum of executive functions studied, but also provides information on the impact of GCR exposure on insightful problem solving, and by default on the functionality of the temporal lobe. • GCR doses as low as 1 cGy impair insightful problem solving. • These data suggest that a wide range of executive functions are impaired by mission pertinent GCR doses, which raises concerns about whether other executive functions that regulate risk decision making and impulsivity are also effected. • The use of PRA style analysis in addition to the traditional cohort analysis, suggest that as many as 2/3 of individuals may experience some form of executive function impairment after 5-10 cGy of Z<15 GCR ions. Brain regions impacted by HZE exposure. The tasks most frequently impaired after GCR exposure were Simple Discrimination (SD) and Compound Discrimination (CD). SD is regulated by the mPFC; CD is probably regulated by perirhinal cortical region, and requires optimal dopamine D2 receptor activation in pre-frontal cortex. Changes in the Neuroproteome associated with changes in ATSET performance. The first batch of pre-frontal cortex samples were run over the Summer of 2017, and Dr. Britten continues to sort through the vast amount of data generated. It is clear that there are proteins whose expression changes as a result of radiation exposure, other proteins whose expression differs in relationship to the ability of the rats to perform ATSET, including GABAergic signaling pathways. Over the remaining period of this grant, we shall analyze more samples from rats exposed to 600 MeV/n 28Si and 400 MeV/n 4He, and given the observed changes in the UCFlex assay we shall be extending these studies to include the Temporal Lobe. Project Summary Over the last year, studies were conducted that determined whether rats exposed to mission-pertinent GCR exposures had compromised cognitive flexibility. The studies have primarily focused on the impact that Z<15 ions had on constrained (ATSET) and unconstrained (creative problem solving) cognitive flexibility. Interestingly, GCR impairment of form of cognitive flexibility was rarely associated with impairment in the other form of cognitive flexibility. These data suggest that risk assessments for cognitive impairment need to be based upon at least two cognitive domains. While some of the observed impairments in executive function could lead to a decreased ability of astronauts to conduct certain tasks, data obtained during this reporting period suggest that "imprinted" memory is not impacted by GCR exposure, so routine procedures will probably be able to be performed during the mission, however, situation requiring novel problem solving could be compromised. Preliminary data suggest that there are marked differences in the neuroproteome in response to GCR exposure and that Beta-adrenergic signaling and GABAergic signaling pathways may be preferentially altered by GCR exposure. Reference Hecht PM, Will MJ, Schachtman TR, Welby LM, Beversdorf DQ. Beta-adrenergic antagonist effects on a novel cognitive flexibility task in rodents. Behav Brain Res, 2014; 260:148-54.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Peer-reviewed Journals	Britten RA, Jewell JS, Duncan VD, Davis LK, Hadley MM, Wyrobek AJ. "Spatial memory performance of socially mature Wistar rats is impaired after exposure to low (5 cGy) doses of 1 GeV/n 48Ti particles." Radiation Research. 2017 Jan;187(1):60-5. https://doi.org/10.1667/RR14550.1 ; PubMed PMID: 28085638 , Jan-2017
Articles in Peer-reviewed Journals	Britten RA, Jewell JS, Davis LK, Miller VD, Hadley MM, Semmes OJ, Lonart G, Dutta SM. "Changes in the hippocampal proteome associated with spatial memory impairment after exposure to low (20 cGy) doses of 1 GeV/n 56Fe radiation." Radiation Research. 2017 Mar;187(3):287-297. Epub 2017 Feb 3. https://doi.org/10.1667/RR14067.1 ; PubMed PMID: 28156212 , Mar-2017
Articles in Peer-reviewed Journals	Dutta SM, Hadley MM, Peterman S, Jewell JS, Duncan VD, Britten RA. "Quantitative proteomic analysis of the hippocampus of rats with GCR-induced spatial memory impairment." Radiation Research. 2017 Dec 5. [Epub ahead of print] https://doi.org/10.1667/RR14822.1 ; PubMed PMID: 29206597 , Dec-2017